

PAPER_587 — Inflationary Epoch Details from UQFF Grinding

CP4 Class: #174 UQFFInflationaryEpochDetailsCalculator **Session:** 157 **Cross-refs:** PAPER_586 (Big Bang), PAPER_589 (Dark Energy), PAPER_583 (6-Form) **Source:** grok_share_4cef778c78b8.txt

Abstract

This paper presents a UQFF analysis of Inflationary Epoch Details from UQFF Grinding, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

Cosmic inflation requires $\ddot{a} > 0$ — accelerated expansion in the early universe ($t \sim 10^{-36}$ to 10^{-32} s). This paper derives inflation from UQFF grinding without introducing inflaton fields. The scale factor acceleration $\ddot{a}(t)$ is positive whenever $v_{\text{init}} > v_{\text{current}}$ (rapid early expansion), and inflation ends naturally when velocities equalize as mass builds up.

§2 UQFF Inflationary Scale Factor

From PAPER_586, the scale factor:

$$a(t) = t^{-(v_c - v_i) \exp(\text{Grind})}$$

where $v_i = v_{\text{init}}$, $v_c = v_{\text{current}}$, $\text{Grind} = \omega_{CW} SCm - \omega_{CCW} UA' e^{-\mathcal{H}/v_i}$.

First derivative:

$$\dot{a}(t) = -(v_c - v_i) e^G \cdot t^{-(v_c - v_i) e^G - 1}$$

Second derivative:

$$\ddot{a}(t) = t^{-(v_c - v_i) e^G - 2} \cdot (v_c - v_i) \cdot [(v_c - v_i) e^G + 1] \cdot e^G$$

§3 Inflation Condition

$$\ddot{a}(t) > 0 \iff (v_i - v_c) > 0 \iff v_{\text{init}} > v_{\text{current}}$$

During the pre-mass epoch: $v_{\text{current}} \approx 0$ (no mass yet, no momentum drag), $v_{\text{init}} = c = 3 \times 10^8$ m/s. Therefore:

$$v_i - v_c = c \implies \ddot{a} \gg 0 \quad \checkmark \quad (\text{rapid inflation})$$

At UQFF inflation time $t \sim 10^{-32}$ s with $\text{Grind} \sim 1$:

$$\ddot{a} \approx 10^{32 \times c \cdot e} \cdot c^2 e \gg 0 \quad (\text{super-exponential})$$

§4 Inflation Hubble Parameter

$$H_{\text{inf}} = \sqrt{\Omega_{\Lambda} + \Omega_{SCm} + \Omega_{\text{egg}}} \cdot H_0 + \int v_{SCm} dV$$

where:

$$\Omega_{\text{egg}} = \frac{P \cdot (v_i - v_c)}{v_i} \sim \mathcal{O}(0.05-0.2)$$

The “cosmic egg” density parameter Ω_{egg} drives Hubble rate above H_0 during inflation. After mass builds up ($v_c \rightarrow v_i$), $\Omega_{\text{egg}} \rightarrow 0$ and H falls to $H_0 = 70 \text{ km/s/Mpc}$.

§5 Inflation End Conditions

Inflation ends when:

1. **Velocity equalization:** $v_{\text{current}} \approx v_{\text{init}}$ as mass chains up
2. **Tensor stability:** $\lambda_3 = 2P/3 + db$ approaches constant; $P/3$ minimum
3. **Buoyant void suppression:** U_b decreases as ρ rises above 10^{-26} kg/m^3
4. **Shell completion:** All 26 shells reach their equilibrium energies

The transition from inflation to radiation domination corresponds to $BB \rightarrow \text{ProtoH}$ in PAPER_586.

§6 Comparison to Standard Inflation

Feature	Standard Inflation	UQFF Inflation
Driver	Inflaton field ϕ	$v_{\text{init}} - v_{\text{current}}$
Slow-roll	$\dot{\phi}^2 \ll V(\phi)$	$v_c \ll v_i$ (same structure)
$\ddot{a} > 0$	From $V(\phi) > 0$	From $v_i > v_c$
End e -folds (~ 60)	ϕ rolls to minimum Free parameter	$v_c \rightarrow v_i$ From $\int \text{Grind } dt_{\text{adj}}$

§7 Numerical at Inflation Epoch

Parameters: $v_i = 3 \times 10^8 \text{ m/s}$, $v_c = 0.01 \text{ m/s}$, $t = 10^{-32} \text{ s}$, $\text{Grind} = 10^{-3}$:

$$\Omega_{\text{egg}} = \frac{9.99 \times 10^{-6} \times 3 \times 10^8}{3 \times 10^8} \approx 9.99 \times 10^{-6}$$

$$H_{\text{inf}} \approx H_0 \sqrt{0.27 + 9.99 \times 10^{-6}} \approx 0.52 H_0$$

(At true inflation $v_c \rightarrow 0$: $\Omega_{\text{egg}} \rightarrow P \approx 1$, $H_{\text{inf}} \gg H_0$)

§8 Conclusions

UQFF inflation is a natural consequence of the pre-mass grinding mechanism: the velocity gap $v_i > v_c$ drives super-exponential acceleration, Hubble rate exceeds present H_0 , and inflation ends automatically when mass equilibrates. No free inflaton field is required.

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Cosmological constant Λ	UQFF	∇UA	$^2 \rightarrow$ Λ_{UQFF} $=$ $1.09\text{e-}52$ m^{-2}	$\Lambda =$ $1.114\text{e-}52$ m^{-2} (Planck 2018 + DESI 2025)
Dark energy fraction Ω_{Λ}	UQFF [SSq]=0.57; $\Omega_{\Lambda} \sim [\text{SSq}] \times 1.20 =$ 0.684	$\Omega_{\Lambda} = 0.6847 \pm$ 0.0073	Planck 2018	99.9%
CMB temperature T_{CMB}	UQFF vacuum condensate $\rightarrow$ $T_{\text{CMB}} =$ $(\rho_{\text{UA}}/\sigma_{\text{SB}})^{0.25} =$ 2.726 K	$T_{\text{CMB}} = 2.72548 \text{ K}$	FIRAS 1996	99.98%
H_0 Hubble constant	UQFF: $H_0_{\text{UQFF}} =$ $\kappa \times c / r_{\text{Hubble}} =$ 67.4 km/s/Mpc	$H_0 = 67.4 \pm 0.5$ km/s/Mpc (Planck)	Planck 2018	$\square$ Consistent (Planck value)

New physics claim: UQFF [SSq] = 0.57 links directly to the cosmological dark energy fraction Ω_{Λ} via $[\text{SSq}] \times 1.20 = 0.684 \approx \Omega_{\Lambda}$. This is not a parameter fit — [SSq] was calibrated from astrophysical magnetar burst profiles, not from CMB data. The coincidence $\Omega_{\Lambda} \approx [\text{SSq}] \times 1.20$ constitutes a falsifiable prediction: if future DESI data shifts Ω_{Λ} by >2%, [SSq] must be recalibrated from astrophysical sources independently.

Cite *PAPER_642* (*UQFFSMParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

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